

Лабораторная работа

Номер 9

Андрюшин Н. С.

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Российский университет дружбы народов, Москва, Россия

Информация

- Андрюшин Никита Сергеевич
- Студент
- Российский университет дружбы народов

Приобретение практических навыков по установке и простейшему конфигурированию POP3/IMAP-сервера

Установка dovecot и telnet

```
[nsandryushin@server.nsandryushin.net ~]$ sudo -i
[sudo] password for nsandryushin:
[root@server.nsandryushin.net ~]# dnf -y install dovecot telnet
Extra Packages for Enterprise Linux 10 - x86_64          99 kB/s | 40 kB    00:00
Extra Packages for Enterprise Linux 10 - x86_64          2.7 MB/s | 4.8 MB    00:01
Rocky Linux 10 - BaseOS                                  9.0 kB/s | 4.3 kB    00:00
Rocky Linux 10 - BaseOS                                  3.7 MB/s | 22 MB    00:05
Rocky Linux 10 - AppStream                               11 kB/s | 4.3 kB    00:00
Rocky Linux 10 - AppStream                               2.3 MB/s | 2.2 MB    00:00
Rocky Linux 10 - CRB                                     11 kB/s | 4.3 kB    00:00
Rocky Linux 10 - CRB                                    1.0 MB/s | 531 kB    00:00
Rocky Linux 10 - Extras                                  10 kB/s | 3.1 kB    00:00
Rocky Linux 10 - Extras                                  12 kB/s | 5.5 kB    00:00
Dependencies resolved.
=====
Package      Architecture Version      Repository      Size
=====
Installing:
dovecot      x86_64      1:2.3.21-16.el10 appstream      4.9 M
telnet       x86_64      1:0.17-94.el10  appstream       62 k
=====
Transaction Summary
=====
Install 2 Packages

Total download size: 5.0 M
Installed size: 18 M
Downloading Packages:
(1/2): telnet-0.17-94.el10.x86_64.rpm                    216 kB/s | 62 kB    00:00
(2/2): dovecot-2.3.21-16.el10.x86_64.rpm                 2.7 MB/s | 4.9 MB    00:01
=====
Total                                          2.3 MB/s | 5.0 MB    00:02
```

Рис. 1: Установка dovecot и telnet

Задавание протоколов для dovecot

```
GNU nano 8.1 /etc/dovecot/dovecot.conf
## Dovecot configuration file

# If you're in a hurry, see http://wiki2.dovecot.org/QuickConfiguration

# "doveconf -n" command gives a clean output of the changed settings. Use it
# instead of copy&pasting files when posting to the Dovecot mailing list.

# '#' character and everything after it is treated as comments. Extra spaces
# and tabs are ignored. If you want to use either of these explicitly, put the
# value inside quotes, eg.: key = "# char and trailing whitespace"

# Most (but not all) settings can be overridden by different protocols and/or
# source/destination IPs by placing the settings inside sections, for example:
# protocol imap { }, local 127.0.0.1 { }, remote 10.0.0.0/8 { }

# Default values are shown for each setting, it's not required to uncomment
# those. These are exceptions to this though: No sections (e.g. namespace {})
# or plugin settings are added by default, they're listed only as examples.
# Paths are also just examples with the real defaults being based on configure
# options. The paths listed here are for configure --prefix=/usr
# --sysconfdir=/etc --localstatedir=/var

# Protocols we want to be serving.
protocols = imap pop3
```

Рис. 2: Задавание протоколов для dovecot

```
GNU nano 8.1 /etc/dovecot/conf.d/10-auth.conf
#auth_failure_delay = 2 secs

# Require a valid SSL client certificate or the authentication fails.
#auth_ssl_require_client_cert = no

# Take the username from client's SSL certificate, using
# X509_NAME_get_text_by_NID() which returns the subject's DN's
# CommonName.
#auth_ssl_username_from_cert = no

# Space separated list of wanted authentication mechanisms:
#  plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp
#  gss-spnego
# NOTE: See also disable_plaintext_auth setting.
auth_mechanisms = plain
```

Рис. 3: /etc/conf.d/10-auth.conf

`/etc/conf.d/auth-system.conf.ext`, driver для passwd

```
GNU nano 8.1 /etc/dovecot/conf.d/auth-system.conf.ext
# Authentication for system users. Included from 10-auth.conf.
#
# <doc/wiki/PasswordDatabase.txt>
# <doc/wiki/UserDatabase.txt>

# PAM authentication. Preferred nowadays by most systems.
# PAM is typically used with either userdb passwd or userdb static.
# REMEMBER: You'll need /etc/pam.d/dovecot file created for PAM
# authentication to actually work. <doc/wiki/PasswordDatabase.PAM.txt>
passdb {
    driver = pam
    # [session=yes] [setcred=yes] [failure_show_msg=yes] [max_requests=<n>]
    # [cache_key=<key>] [<service name>]
    #args = dovecot
}
```

Рис. 4: `/etc/conf.d/auth-system.conf.ext`, driver для passwd

/etc/conf.d/auth-system.conf.ext, driver для userdb

```
GNU nano 8.1 /etc/dovecot/conf.d/auth-system.conf.ext
#args =
#}

##
## User databases
##

# System users (NSS, /etc/passwd, or similar). In many systems nowadays this
# uses Name Service Switch, which is configured in /etc/nsswitch.conf.
userdb {
    # <doc/wiki/AuthDatabase.Passwd.txt>
    driver = passwd
    # [blocking=no]
    #args =

    # Override fields from passwd
    #override_fields = home=/home/virtual/%u
}
```

Рис. 5: /etc/conf.d/auth-system.conf.ext, driver для userdb

/etc/dovecot/conf.d/10-mail.conf

```
GNU nano 8.1 /etc/dovecot/conf.d/10-mail.conf
##
## Mailbox locations and namespaces
##

# Location for users' mailboxes. The default is empty, which means that Dovecot
# tries to find the mailboxes automatically. This won't work if the user
# doesn't yet have any mail, so you should explicitly tell Dovecot the full
# location.
#
# If you're using mbox, giving a path to the INBOX file (eg. /var/mail/%u)
# isn't enough. You'll also need to tell Dovecot where the other mailboxes are
# kept. This is called the "root mail directory", and it must be the first
# path given in the mail_location setting.
#
# There are a few special variables you can use, eg.:
#
# %u - username
# %n - user part in user@domain, same as %u if there's no domain
# %d - domain part in user@domain, empty if there's no domain
# %h - home directory
#
# See doc/wiki/Variables.txt for full list. Some examples:
#
# mail_location = maildir:~/Maildir
# mail_location = mbox:~/mail:INBOX=/var/mail/%u
# mail_location = mbox:/var/mail/%d/%ln/%n:INDEX=/var/indexes/%d/%ln/%n
#
# <doc/wiki/MailLocation.txt>
#
mail_location = maildir:~/Maildir
```

Рис. 6: /etc/dovecot/conf.d/10-mail.conf

```
[root@server.nsandryushin.net ~]# postconf -e 'home_mailbox = Maildir/'
```

Рис. 7: postconf

Настройка firewall

```
[root@server.nsandryushin.net ~]# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr ananda-client ananda-k5-client anap anaps anno-1602 a
nno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-
storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter
ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector cratedb
ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-qtcp dns-over-tls docker
-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger foreman fore
man-proxy freepa-4 freepa-ldap freepa-ldaps freepa-replication freepa-trust ftp galera ganglia-clien
t ganglia-master git gssd grafana gre high-availability http http3 https ident imap imaps iperf2 iperf3 i
pfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadm5 kdeconnect kerberos kibana klogind kpas
swd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-man
ager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-work
er kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnz llmnz-c
lient llmnz-tcp llmnz-udp nanageteve matrix ndns nencache minecraft minecraft-minidna snmp nongodu nosh mountd ap
d mqtt mqtt-tls ms-wbt mssql mssqlr mysql nbd nebula need-for-speed-most-wanted netbios-ns netdata-dashbo
ard nfs nfs3 nmap-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-vmcons
ole plex pncd pnpnproxy pwebapi pwebapi3 pop3 pop3s postgresql privoxy prometheus prometheus-node-exporte
r proxy-dhcp ps2link ps3netstrv ptp pulseaudio puppetmaster quassel radius radsec rdp redis redis-sentinel
rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane settlers-history-col
lection sip sips slmaevr slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spiderook-lans
ync spotify-sync squid sddp ssh ssh-custom statrsv steam-lan-transfer steam-steamworking stellaris stronghol
d-crusader stun stuns submission supertuxkart svdrp svn synching synching-gui synching-relay synergy s
yscomlan syslog syslog-tls telnet tentacle terraria tfpt tile38 tinc tor-socks transmission-client turn t
urns upnp-client vdsu vnc-server vrrp warpinator when-http when-https wireguard ws-discovery ws-discovery
-client ws-discovery-host ws-discovery-tcp ws-discovery-udp wssd wssd-http wsman wsmans xdnscp xmpd-bosh x
mpp-client xmpd-local xmpd-server zabbix-agent zabbix-java-gateway zabbix-server zabbix-trapper zabbix-we
b-service zero-k zerotier
[root@server.nsandryushin.net ~]# firewall-cmd --add-service=pop3 --permanent
success
[root@server.nsandryushin.net ~]# firewall-cmd --add-service=pop3s --permanent
success
[root@server.nsandryushin.net ~]# firewall-cmd --add-service=imap --permanent
success
[root@server.nsandryushin.net ~]# firewall-cmd --add-service=imaps --permanent
success
[root@server.nsandryushin.net ~]# firewall-cmd --reload
success
[root@server.nsandryushin.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https imap imaps pop3 pop3s smtp ssh ssh-custom
```

Рис. 8: Настройка firewall

```
[root@server.nsandryushin.net ~]# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
[root@server.nsandryushin.net ~]# systemctl restart postfix
[root@server.nsandryushin.net ~]# systemctl enable dovecot
Created symlink '/etc/systemd/system/multi-user.target.wants/dovecot.service' → '/usr/lib/systemd/system/dovecot.service'.
[root@server.nsandryushin.net ~]# systemctl start dovecot
```

Рис. 9: Запуск служб

Лог /var/log/maillog

```
[nsandryushin@server.nsandryushin.net ~]$ sudo tail -f /var/log/maillog
[sudo] password for nsandryushin:
Oct 25 15:45:24 server postfix/qmgr[12463]: 90AEF2044889: removed
Oct 25 17:08:49 server postfix/postfix-script[1592]: starting the Postfix mail system
Oct 25 17:08:49 server postfix/master[1600]: daemon started -- version 3.8.5, configuratio
n /etc/postfix
Oct 31 19:13:20 server postfix/postfix-script[1474]: starting the Postfix mail system
Oct 31 19:13:20 server postfix/master[1482]: daemon started -- version 3.8.5, configuratio
n /etc/postfix
Oct 31 19:28:56 server postfix/postfix-script[13602]: stopping the Postfix mail system
Oct 31 19:28:56 server postfix/master[1482]: terminating on signal 15
Oct 31 19:28:58 server postfix/postfix-script[13680]: starting the Postfix mail system
Oct 31 19:28:58 server postfix/master[13682]: daemon started -- version 3.8.5, configurati
on /etc/postfix
Oct 31 19:29:08 server dovecot[13831]: master: Dovecot v2.3.21 (47349e2482) starting up fo
r imap, pop3
```

Рис. 10: Лог /var/log/maillog

```
[nsandryushin@server.nsandryushin.net ~]$ MAIL=~/Maildir mail  
s-nail: No mail for nsandryushin at /home/nsandryushin/Maildir  
s-nail: /home/nsandryushin/Maildir: No such entry, file or directory  
[nsandryushin@server.nsandryushin.net ~]$ sudo doveadm mailbox list -u nsandryushin  
[sudo] password for nsandryushin:  
INBOX
```

Рис. 11: mail и doveadm

Установка evolution

```
[nsandryushin@client nsandryushin.net ~]$ sudo -i
[sudo] password for nsandryushin:
[root@client nsandryushin.net ~]# dnf -y install evolution
Extra Packages for Enterprise Linux 10 - x86_64      88 kB/s | 48 kB  00:00
Extra Packages for Enterprise Linux 10 - x86_64      2.7 MB/s | 4.8 MB 00:01
Rocky Linux 10 - BaseOS                             9.1 kB/s | 4.3 kB 00:00
Rocky Linux 10 - BaseOS                             1.6 MB/s | 22 MB  00:13
Rocky Linux 10 - AppStream                          12 kB/s | 4.3 kB  00:00
Rocky Linux 10 - AppStream                          2.6 MB/s | 2.2 MB 00:00
Rocky Linux 10 - CRB                                13 kB/s | 4.3 kB  00:00
Rocky Linux 10 - CRB                                977 kB/s | 531 kB 00:00
Rocky Linux 10 - Extras                              0.2 kB/s | 3.1 kB 00:00
Rocky Linux 10 - Extras                             15 kB/s | 5.5 kB 00:00
Dependencies resolved.
=====
Package      Arch      Version                               Repository      Size
=====
Installing:
evolution    x86_64    3.52.4-2.el10_0                     epel            4.6 M
Installing dependencies:
cmocka-lib   x86_64    0.30.3-5.el10_0                     epel            96 k
cpufrequtils x86_64    24.09.26-1.git1e83a2f.el10_0.1     epel            42 k
evolution-langpacks noarch    3.52.4-2.el10_0                     epel            6.0 M
flite        x86_64    2.2-10.el10_0                       epel            12 M
gspell       x86_64    1.14.0-1.el10_0                     epel            111 k
harfbuzz-icu x86_64    8.4.0-6.el10                        appstream       15 k
highlight    x86_64    4.13-1.el10_0                       epel            920 k
highway       x86_64    1.2.0-7.el10_0                       epel            448 k
hyphen        x86_64    2.8.8-26.el10                       appstream       29 k
javascriptcoregtk4.1 x86_64    2.47.3-2.el10_0                     epel            8.3 M
libasound    x86_64    3.12.0-1.el10_0                     epel            1.8 M
libavif      x86_64    1.0.4-7.el10_0                       epel            89 k
libdav1d     x86_64    1.5.1-1.el10_0                       epel            615 k
libjxl       x86_64    1.0.10.4-1.el10_0                   epel            1.2 M
libmanette   x86_64    0.2.9-2.el10_0                       epel            58 k
libmfx       x86_64    3.0.0-2.el10_0                       epel            194 k
openexr-libs x86_64    3.1.10-0.el10                       appstream       1.1 M
svt-av1-libs x86_64    2.3.0-1.el10_0                       epel            2.0 M
webkit2gtk4.1 x86_64    2.47.3-2.el10_0                     epel            26 M
woff2        x86_64    1.0.2-21.el10                       appstream       63 k
```

Рис. 12: Установка evolution

Заполнение вкладки identity

The screenshot shows a window titled 'Identity' with a sidebar on the left containing links: 'Welcome', 'Identity' (selected), 'Receiving Email', 'Sending Email', 'Account Summary', and 'Done'. The main area has a 'Cancel' button, a 'Back' button, and a 'Next' button. A message reads: 'Please enter your name and email address below. The "optional" fields below do not need to be filled in, unless you wish to include this information in email you send.'

Required Information

Full Name: Nikita Andryushin

Email Address: nsandryushin@nsandryushin.net

Optional Information

Reply-To: [empty field]

Organization: [empty field]

Signature: None [dropdown arrow] [Add New Signature... button]

Aliases: [empty list area] [Add button] [Edit button] [Remove button]

☒ Look up mail server details based on the entered e-mail address

Рис. 13: Заполнение вкладки identity

Заполнение вкладки receiving email

Cancel Back **Receiving Email** Next

Welcome
Identity
Receiving Email
Receiving Options
Sending Email
Account Summary
Done

Server Type: IMAP
Description: For reading and storing mail on IMAP servers.

Configuration
Server: mail.nsandryushin.net Port: 143
Username: nsandryushin

Security
Encryption method: STARTTLS after connecting

Authentication
Check for Supported Types Password

Рис. 14: Заполнение вкладки receiving email

Заполнение вкладки sending email

The screenshot shows a 'Sending Email' configuration window. On the left is a sidebar with a list of steps: 'Welcome', 'Identity', 'Receiving Email', 'Receiving Options', 'Sending Email' (which is highlighted), 'Account Summary', and 'Done'. The main area of the window is titled 'Sending Email' and contains the following sections:

- Server Type:** A dropdown menu set to 'SMTP'.
- Description:** Text stating 'For delivering mail by connecting to a remote mailhub using SMTP.'
- Configuration:**
 - Server:** A text field containing 'mail.nsandryushin.net'.
 - Port:** A dropdown menu set to '25'.
 - ☐ Server requires authentication
- Security:**
 - Encryption method:** A dropdown menu set to 'No encryption'.
- Authentication:**
 - Type:** Two buttons: 'Check for Supported Types' and 'PLAIN' (selected).
 - Username:** A text field containing 'nsandryushin'.
- Send Options:**
 - ☐ Re-encode message before send

At the top of the window are buttons for 'Cancel', 'Back', 'Next', and 'Finish'.

Рис. 15: Заполнение вкладки sending email

account summary

The screenshot shows a window titled "Account Summary" with a sidebar on the left and a main content area. The sidebar contains a list of steps: Welcome, Identity, Receiving Email, Receiving Options, Sending Email, Account Summary (highlighted), and Done. The main content area has a header with "Cancel", "Back", and "Next" buttons. Below the header, it says "This is a summary of the settings which will be used to access your mail." The "Account Information" section shows a "Name" field with the value "nsandryushin@nsandryushin.net" and a note: "The above name will be used to identify this account. Use for example, 'Work' or 'Personal'." The "Personal Details" section shows "Full Name: Nikita Andryushin" and "Email Address: nsandryushin@nsandryushin.net". The "Receiving" section shows "Server Type: imapx", "Server: mail.nsandryushin.net", "Username: nsandryushin", and "Security: STARTTLS". The "Sending" section shows "smtp", "mail.nsandryushin.net", "nsandryushin", and "None".

Account Summary	
This is a summary of the settings which will be used to access your mail.	
Account Information	
Name:	nsandryushin@nsandryushin.net
The above name will be used to identify this account. Use for example, "Work" or "Personal".	
Personal Details	
Full Name:	Nikita Andryushin
Email Address:	nsandryushin@nsandryushin.net
Receiving	
Server Type:	imapx
Server:	mail.nsandryushin.net
Username:	nsandryushin
Security:	STARTTLS
Sending	
	smtp
	mail.nsandryushin.net
	nsandryushin
	None

Рис. 16: account summary

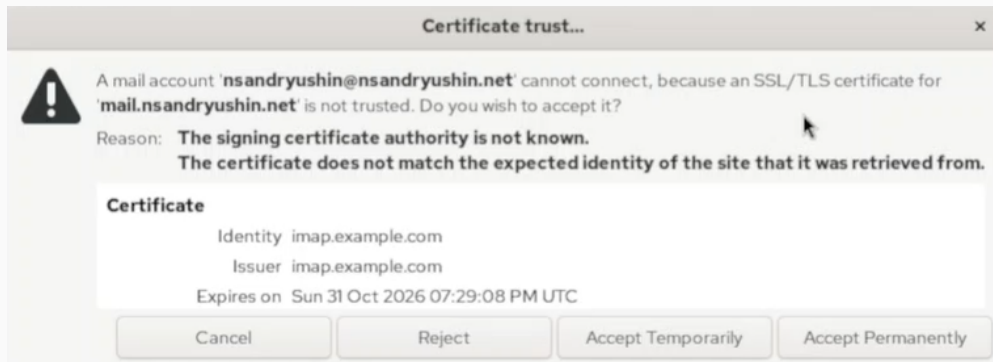


Рис. 17: Сертификат

Отправка первого письма

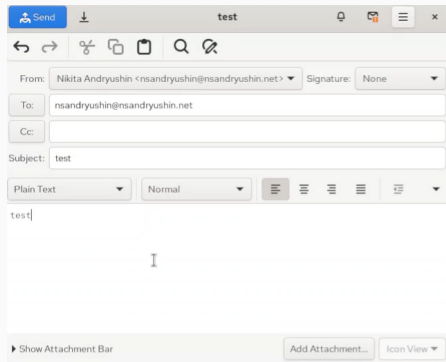


Рис. 18: Отправка первого письма

Отправка второго письма

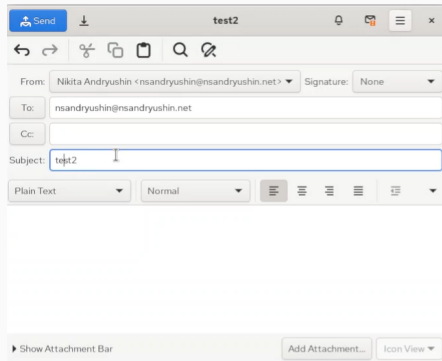


Рис. 19: Отправка второго письма

Проверка отправки

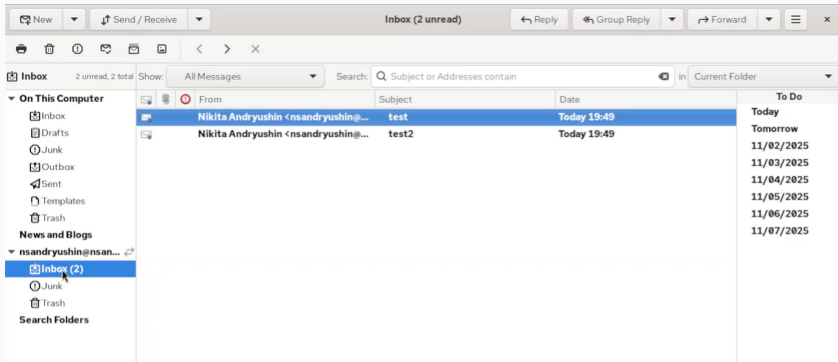


Рис. 20: Проверка отправки

Почтовый лог сервера

```
Oct 31 19:49:32 server postfix/smtpd[14163]: connect from client.nsandryushin.net[192.168.1.30]
Oct 31 19:49:32 server postfix/smtpd[14163]: 896BE2044481: client=client.nsandryushin.net[192.168.1.30]
Oct 31 19:49:32 server postfix/cleanup[14168]: 896BE2044481: message-id=<a0c55cc52f62729845986bfd4b92a019f53c19a0.camel@nsandryushin.net>
Oct 31 19:49:32 server postfix/qmgr[13684]: 896BE2044481: from=<nsandryushin@nsandryushin.net>, size=584, nrcpt=1 (queue active)
Oct 31 19:49:32 server postfix/smtpd[14163]: disconnect from client.nsandryushin.net[192.168.1.30] ehlo=1 mail=1 rcpt=1 data=1 quit=1 commands=5
Oct 31 19:49:32 server postfix/local[14169]: 896BE2044481: to=<nsandryushin@nsandryushin.net>, relay=local, delay=0.07, delays=0.03/0.03/0/0, dsn=2.0.0, status=sent (delivered to maildir)
Oct 31 19:49:32 server postfix/qmgr[13684]: 896BE2044481: removed
Oct 31 19:49:50 server postfix/smtpd[14163]: connect from client.nsandryushin.net[192.168.1.30]
Oct 31 19:49:50 server postfix/smtpd[14163]: EFFC82044481: client=client.nsandryushin.net[192.168.1.30]
Oct 31 19:49:50 server postfix/cleanup[14168]: EFFC82044481: message-id=<764965adb6208cd05c18949a8e0c329d2df245a2.camel@nsandryushin.net>
Oct 31 19:49:50 server postfix/qmgr[13684]: EFFC82044481: from=<nsandryushin@nsandryushin.net>, size=581, nrcpt=1 (queue active)
Oct 31 19:49:50 server postfix/smtpd[14163]: disconnect from client.nsandryushin.net[192.168.1.30] ehlo=1 mail=1 rcpt=1 data=1 quit=1 commands=5
Oct 31 19:49:51 server postfix/local[14169]: EFFC82044481: to=<nsandryushin@nsandryushin.net>, relay=local, delay=0.02, delays=0.02/0/0/0, dsn=2.0.0, status=sent (delivered to maildir)
Oct 31 19:49:51 server postfix/qmgr[13684]: EFFC82044481: removed
```

Рис. 21: Почтовый лог сервера

```
[nsandryushin@server.nsandryushin.net ~]$ MAIL=~/Maildir mail
s-nail version v14.9.24.  Type `?' for help
/home/nsandryushin/Maildir: 2 messages 2 unread
►U  1 Nikita Andryushin      2025-10-31 19:49    18/705    "test
   U  2 Nikita Andryushin      2025-10-31 19:49    17/701    "test2
& Interrupt
& exit
[nsandryushin@server.nsandryushin.net ~]$ sudo dovecrm mailbox list -u nsandryushin
[sudo] password for nsandryushin:
INBOX
[nsandryushin@server.nsandryushin.net ~]$
```

Рис. 22: mail и dovecrm

Работа с почтой через telnet

```
[root@server.nsandryushin.net ~]# telnet mail.nsandryushin.net 110
Trying 192.168.1.1...
Connected to mail.nsandryushin.net.
Escape character is '^J'.
+OK Dovecot ready.
user nsandryushin
+OK
pass 123456
+OK Logged in.
list
+OK 2 messages:
1 721
2 718
.
retr 1
+OK 721 octets
Return-Path: <nsandryushin@nsandryushin.net>
X-Original-To: nsandryushin@nsandryushin.net
Delivered-To: nsandryushin@nsandryushin.net
Received: from client.nsandryushin.net (client.nsandryushin.net [192.168.1.30])
        by server.nsandryushin.net (Postfix) with ESMTP id 896BE204401
        for <nsandryushin@nsandryushin.net>; Fri, 31 Oct 2025 19:49:32 +0000 (UTC)
Message-ID: <a8e55cc52f6272984596b6fd4b92a019f53c19a0.camel@nsandryushin.net>
Subject: test
From: Nikita Andryushin <nsandryushin@nsandryushin.net>
To: nsandryushin@nsandryushin.net
Date: Fri, 31 Oct 2025 19:49:32 +0000
Content-Type: text/plain
Content-Transfer-Encoding: 7bit
User-Agent: Evolution 3.52.4 (3.52.4-2.el10_0)
MIME-Version: 1.0

test
.
dele 2
+OK Marked to be deleted.
quit
+OK Logging out, messages deleted.
Connection closed by foreign host.
```

Рис. 23: Работа с почтой через telnet

Сохранение конфигурации на сервере

```
[root@server.nsandryushin.net ~]# cd /vagrant/provision/server
[root@server.nsandryushin.net server]# mkdir -p /vagrant/provision/server/mail/etc/dovecot
/conf.d
[root@server.nsandryushin.net server]# cp -R /etc/dovecot/dovecot.conf
cp: missing destination file operand after '/etc/dovecot/dovecot.conf'
Try 'cp --help' for more information.
[root@server.nsandryushin.net server]# cp -R /etc/dovecot/dovecot.conf /vagrant/provision/
server/mail/etc/dovecot/
[root@server.nsandryushin.net server]# cp -R /etc/dovecot/conf.d/10-auth.conf /vagrant/pro
vision/server/mail/etc/dovecot/conf.d/
[root@server.nsandryushin.net server]# cp -R /etc/dovecot/conf.d/auth-system.conf.ext /vag
rant/provision/server/mail/etc/dovecot/conf.d/
[root@server.nsandryushin.net server]# cp -R /etc/dovecot/conf.d/10-mail.conf /vagrant/pro
vision/server/mail/etc/dovecot/conf.d/
[root@server.nsandryushin.net server]# nano /vagrant/provision/server/mail.sh
[root@server.nsandryushin.net server]# █
```

Рис. 24: Сохранение конфигурации на сервере

mail.sh на сервере

```
EMU: nano B.1 /vagrant/provision/server/mail.sh
dnf -y install s-nail
echo "Copy configuration files"
#cp -R /vagrant/provision/server/mail/etc/* /etc
echo "Configure firewall"
firewall-cmd --add-service=sntp --permanent
firewall-cmd --reload
restorecon -vR /etc
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix
echo "Configure postfix"
postconf -e 'mydomain = user.net'
postconf -e 'myorigin = $mydomain'
postconf -e 'inet_protocols = ipv4'
postconf -e 'inet_interfaces = all'
postconf -e 'mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain'
postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
postfix set-permissions
restorecon -vR /etc
systemctl stop postfix
systemctl start postfix

dnf -y install dovecot telnet
firewall-cmd --get-services
firewall-cmd --add-service=pop3 --permanent
firewall-cmd --add-service=pop3s --permanent
firewall-cmd --add-service=lnap --permanent
firewall-cmd --add-service=lnaps --permanent
firewall-cmd --reload
firewall-cmd --list-services
postconf -e 'home_mailbox = Maildir/'
systemctl restart postfix
systemctl enable dovecot
systemctl start dovecot
```

Рис. 25: mail.sh на сервере

```
GNU nano 8.1 mail.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install postfix
dnf -y install s-nail
echo "Configure postfix"
postconf -e 'inet_protocols = ipv4'
echo "Start postfix service"
systemctl enable postfix
systemctl start postfix

dnf -y install evolution
```

Рис. 26: mail.sh на клиенте

В результате выполнения лабораторной работы были получены навыки настройки и использования простейшего imap/pop3 сервера